

Downscaling

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This report is generated for **IND** for the pathway **CurrentTrends**.

Checking input data

Below is the table of input data used in the prior module of the downscaling process.

Variable in econometric model	Year	Unit	Source
Dependent			
Land use change	2015-2019	% of lu.from in pixel being converted	HILDA
Explanatory variables			
Land use	2015	hectares	COPERNICUS
Forest Management	2015	hectares	Lesiv et al.
Mean Travel time to major cities	2000	minutes	Nelson, A (2008)
Population	2020	Count	WorldPop

Checking the land cover change input data

Check in the table below that all land cover of interest and all possible transitions are included in the land transition matrix.

- The structure of the matrix should be such that you have the same rownames and colnames. Check that you have all land covers, by default it should be cropland, forest, new forest, otherland, pasture and urban.
- The matrix should be filled with the same number in each cell (i.e. the number of pixels for your country's grid representation in 50km x 50km)

```
##
##      cropland forest newforest otherland pasture urban
## cropland    1217   1217      1217      1217   1217  1217
## forest      1217   1217      1217      1217   1217  1217
## newforest    1217   1217      1217      1217   1217  1217
## otherland    1217   1217      1217      1217   1217  1217
## pasture      1217   1217      1217      1217   1217  1217
## urban        1217   1217      1217      1217   1217  1217
```

Harmonizing start map with FABLE Calculator baseline map

We need to harmonize the high-resolution baseline map used in the `downscalR` function with the aggregated land-use data from the FABLE Calculator. This ensures that the spatially explicit baseline map aligns with the aggregated land-use data in FABLE. Without this alignment, there could be discrepancies in the starting levels of each land-use class, making the future land-use transitions projected by the FABLE Calculator unrealistic when applied to the unadjusted high-resolution map.

```
## Warning in left_join(start_map, CLC_FABLE_shares): Detected an unexpected many-to-many relationship between
## i Row 1 of 'x' matches multiple rows in 'y'.
## i Row 1 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
## "many-to-many"' to silence this warning.
```

Checking FABLE land use change projections to downscale

Check in the table below that in the land use changes projected in your calculator, all land cover of interest and all transitions are included in the land transition matrix.

- The structure of the matrix should be such that you have the same rownames and colnames. Check that you have all land covers, by default it should be cropland, forest, new forest, otherland, pasture and urban.
- The matrix should be filled with 10 in each cell (i.e. the number of year periods in the calculator).
- The diagonal should be filled with zeros.

```
##
##      cropland forest newforest otherland pasture urban
## cropland      0     10       10       10       10     10
## forest       10      0       10       10       10     10
## newforest    10     10        0       10       10     10
## otherland    10     10       10        0       10     10
## pasture     10     10       10       10        0     10
## urban       10     10       10       10       10      0
```

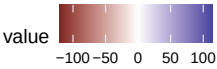
Estimate MNL model

Downscaling

Land-use map



Land-use change map



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